



Refining a model for sustained usage of agile methodologies



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ABSTRACT

This paper refines a model of Sustained Agile Usage to present a comprehensive understanding of the key factors that are pertinent to the sustained usage of agile methodologies. It describes our qualitative study which involves: (i) a focus group with twenty-nine software industry agile practitioners, and (ii) semi-structured interviews with twenty agile practitioners from five different organizational backgrounds. Data from both methods is used to develop the refined model of Sustained Agile Usage. The refined Sustained Agile Usage Model includes the following three categories: Agile Team Factors, Technological Factors, and Organizational Factors. These revisions are discussed in this research. Finally, we see implications for research: the study offers a useful complement to the few studies that have examined the long-term acceptance of agile methods. The refined model can be used as a reference model to guide future studies to understand sustained usage in different agile domains (e.g. Kanban). Additionally, implications for practice include valuable insights that can help agile teams and others (e.g. top management) to better understand and benchmark how agile methods can be effectively sustained in organizations.

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1. Introduction

In recent years, the use of agile method(s) (AM) in information systems development (ISD) has gained significant momentum and is now generally considered a viable approach both in academic and practitioner settings. However, while AMs continue to evolve with newer ones emerging (Williams, 2012), there is a need for better understanding their actual use beyond the adoption stage (Abrahamsson et al., 2009). Prior studies on AMs have focused on their adoption or adaptation in organizational environments, paying more attention to examining factors that drive organizations to initially adopt AMs rather than on those that influence their sustained use (Overhage & Schlauderer, 2012a). In a recent study that maps a landscape of the current challenges practitioners face, Gregory et al., (2016, p. 104) show that "practitioners are less concerned about adopting agile and more concerned about sustaining agile. The sustainability of agile has not been widely researched". Though there is an increasing number of empirical studies on AMs in real-world contexts, the research on AMs in practice is yet to yield significant systematic and insightful knowledge that can pro-

vide credible advice to organizations for maintaining AM use and guide future research (Wang et al., 2012). Two issues represent the main challenges to this branch of research:

- (i) Because agile systems development research is primarily practitioner driven (Goh et al., 2013), a key challenge has been to use a sound theoretical basis to understand the continued and effective use of AMs by exploring the degree or extent of agile practice use (Wang et al., 2012), and
- (ii) Superficial judgement of agile method use as used or not used is assessed without evaluating the effectiveness of their actual implementation (Conboy and Fitzgerald, 2010).

In an attempt to address these gaps in the literature, Senapathi and Srinivasan (2012) developed an initial model of Sustained Agile Usage to explore the key factors that influence the sustained usage of AMs. The model draws from well-established theories such as innovation assimilation, implementation models, and systems development methodology literature. The initial model is a useful lens for illustrating the factors that are critical to sustained agile usage. However, as it is one of the earliest models to conceptualize the phenomenon of sustained agile usage, it was deemed important that the initial model be further refined using empirical analyses. Moreover, given the growing popularity of AMs and their use in many more contexts than initially expected, the aim of this research is to refine the model with a broader set

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of practitioners to ensure we have identified the key factors impacting sustained agile usage. Therefore, the purpose of the current study is to use a multi-method approach to refine the constructs, including the categories and corresponding factors of each category, developed in the initial model in order to gain a deeper understanding of sustained use of AMs. It is believed that such constant comparison across different types of evidence and literature has the ability to generate less biased theory (Eisenhardt, 1989).

Building and extending on Senapathi and Srinivasan (2012)'s study, this current research describes how we refine the initial model as we report on the development of a refined theoretical model of Sustained Agile Usage. The primary objective of this research is to gain a deeper understanding of the factors that influence sustained agile usage. As such we have focused our efforts towards this objective. Other elements of the model (e.g. sustained agile usage factors) have been previously refined in the earlier phases of this research (Senapathi et al., 2011; Senapathi and Srinivasan, 2012). We describe how we used two methods, a focus group and semi-structured interviews, to further explore and gain insights on the adequacy and nature of the theoretically developed factors that influence sustained agile usage.

The structure of the paper is as follows. In the next section we discuss the related literature and basic terminology to distinguish the related concepts of factors that impact the later stages of innovation use. In our case, these later stages refer to sustained agile usage. We then present the initial sustained agile usage model (SAUM). Following this, we discuss our research method and discuss the analysis and findings. We explain the subsequent modifications made to the SAUM and present a summary of the refined factors and the refined SAUM. The overall conclusions and implications are presented. We also present limitations of the study and discuss areas of future research.

2. Theoretical background

Sustained usage refers to the continued use of an innovation, which focuses on the extent and intensity of use beyond the simplistic binary perspective of initial adoption when something is either adopted or not. Emphasizing the significance of continued usage over initial usage, it is argued that the ultimate success of an innovation can only be derived from its sustained usage (Bhattacharjee, 2001). In other words, in order to fully derive the desired benefits and improvements in systems development outcomes, their use needs to be institutionalized as part of the user's regular work behaviours. More recently, research into post-adoptive usage has been receiving greater attention (Jasperson et al., 2005), including agile methods (Mangalaraj et al., 2009; Wang et al., 2012). This current study seeks to contribute to this emerging body of post-adoptive usage literature.

2.1. Sustained or continued use of innovations

The use of systems development methodology (SDM)s in general is a versatile concept (Iivari and Huisman, 2007). Given the lack of strong theoretical and conceptual bases in agile research, the initial SAUM (Senapathi and Srinivasan, 2012) draws from traditional innovation diffusion models (e.g. diffusion of innovations, information system implementation), which have been tested extensively in the information systems (IS) literature (Cooper and Zmud, 1990; Gallivan, 2001; Kwon and Zmud, 1987). Recently, they have generated interest in the agile research community (Schlauderer and Overhage, 2013; Schlauderer et al., 2015; Wang et al., 2012).

The six stages in the traditional innovation models are *initiation*, *adoption*, *adaptation*, *acceptance*, *routinization*, and *infusion* phases (Kwon and Zmud, 1987). While the initial three phases (*initiation*, *adoption*, *adaptation*) relate to adoptive behavior of an innovation, the last three phases (*acceptance*, *routinization*, *infusion*) relate to post-adoptive, or sustained, usage of an innovation. *Acceptance* refers to the commitment of use, while *routinization* is when the usage of the innovation is encouraged as a normal activity. *Infusion* refers to the innovation penetrating deeply into an organization, which is further described using three facets: 1) extensive use: extent of use of an innovation to accommodate a more comprehensive set of work tasks; 2) integrated use: using the innovation to establish or enhance work flow linkages among work tasks; and 3) emergent use: using the innovation beyond its original intended scope (Hsieh and Zmud, 2006; Saga and Zmud, 1994). Adapting it to the context of agile practice assimilation, Wang et al. added two additional indicators, i.e. intensive use and deeply customized use (Wang et al., 2012). While intensive use refers to the use of an agile practice beyond that suggested by the textbook, deeply customized use is the deep adaptation of an agile practice that reflects the needs of the adopting unit/team. For example, an agile team adapting the forty-hour week to align with client working hours (clients work to a split-shift rota) and alternating between 6 and 10 h days, or 4 and 12 h shifts, instead of the recommended 8 hours per day (Wang et al., 2012).

Drawing from the theoretical concepts described above, Senapathi and Srinivasan (2012) define *Sustained Agile Usage* as the combination of *vertical usage* and *horizontal usage*. *Vertical usage* refers to the depth and intensity of use of agile practices of a specific agile method, incorporating one or more combinations of usage behaviours specified in the three infusion facets described above. *Horizontal usage* is concerned with the use of the innovation across the organization, such as percentage of projects and number of teams using the innovation. This includes, for example, the spread of the use of Scrum practices from one team/project to multiple teams/projects, from one region to many regions within an organization, and to other departments (e.g. business, finance). As both the vertical and horizontal usage constructs have been previously refined in the earlier phases of this research related to the development of the SAUM (Senapathi et al., 2011; Senapathi and Srinivasan, 2012), the focus of this study are the categories and the factors that influence sustained agile usage.

2.2. Success factors of the later stages of agile use

Studies that have focused on identifying success factors related to the use of AMs include those conducted by Chan and Thong (2009), Misra et al. (2009), and Chow and Cao (2008). While Chow and Cao (2008) and Misra et al. (2009) focus on success factors relating to agile adoption, Chan and Thong (2009) develop a framework for agile acceptance from a knowledge management perspective. However, the use of the term *acceptance* in Chan and Thong (2009) implies more of an adoption perspective and does not differentiate between adoptive and post-adoptive stages of diffusion.

Few studies (e.g. Mangalaraj et al., (2009) and Wang et al., (2012)) have focused on later stages of agile assimilation: Wang et al. (2012) focused on the assimilation of XP/Scrum practices using three case studies, and Mangalaraj et al. (2009) investigated the acceptance of XP practices across different teams within the same organization. Wang et al. (2012) focused on the later post-adoptive stages, i.e. acceptance, routinization and infusion, contending the need to examine AMs at the practice level. However, Mangalaraj et al. (2009) examined agile acceptance at the method level in a broad sense without differentiating it from the other two post-stages, i.e. routinization, and in-

fusion (Wang et al., 2012). By mapping specific Scrum benefit characteristics, such as time-to-market, collaboration, and transparency to Diffusion of Innovation (DOI) factors (i.e. relative advantage, compatibility and complexity factors), Overhage and Schlauderer (2012a) examine the long-term acceptance of Scrum in a large organization where the focus is limited to innovation factors. Schlauderer et al. (2015) examine the acceptance of Water-Scrum-Fall (i.e. Scrum combined with waterfall) projects, while Mohan et al. (2013) conceptualize the usage of SDMs using three distinct factors: committed, compliant, and resistant use. They also examine how DOI attributes such as *relative advantage* and *compatibility* influence usage behaviour.

The above studies have contributed to building a stronger theoretical base in AMs and gaining an initial understanding of acceptance factors of agile use. While this initial understanding of acceptance factors is an important first step towards realizing the successful use of AMs, their long-term viability and eventual effectiveness depend on their continued and sustained use. Further, prior research does not elaborate on practitioners' motivations emerging after their initial acceptance, which potentially influence their subsequent usage decisions but not their initial acceptance. As a result, some major gaps still exist in this stream of literature:

- The significance of investigating the barriers and facilitators of sustained usage that have not been explored (Mangalaraj et al., 2009; Wang et al., 2012)
- Most studies have used only innovation factors to investigate acceptance and ignored other important factors, particularly given the fact that AMs place more emphasis on 'people' (sociological and organizational factors) (Cockburn & Highsmith, 2001)
- A focus on "sustainability" or "long-term acceptance" without actual conceptualization or measurement of "acceptance" as a post-adoptive phase
- A focus on individual methods such as Scrum (Overhage & Schlauderer, 2012b) and XP (Mangalaraj et al., 2009; Wang et al., 2012) has resulted in fragmented research by missing the commonalities of "agile" methods (livari and livari, 2011)

Since it might be intellectually economic to focus on AMs as a whole (livari and livari, 2011), the current study seeks to cover the totality of agile methods rather than on individual methods or practices. In response to these gaps, Senapathi and Srinivasan (2012) conceptualized a model of Sustained Agile Usage taking into account the distinction between acceptance and continued use, i.e. sustained use. Building and extending on Senapathi and Srinivasan (2012)'s study, our research describes how we refine the initial model as we report on the development of a refined theoretical model of Sustained Agile Usage. The objective is to develop a refined set of theoretical factors that contribute to the phenomenon of sustained agile usage, and to offer these concepts to both the research and practitioner community to apply to other cases. The qualitative multi-method approach employed here provides rich data to refine the initial model.

3. Initial sustained agile usage model

We present the initial sustained agile usage model in Fig. 1. The left hand side of the vertical dotted lines represents the four categories and their factors that influence sustained agile usage and the right hand side represents the consequences or outcomes of sustained agile usage.

3.1. Sustainability factors that impact sustained agile usage

Understanding the various factors that impact usage has generated much interest in IS research (Agarwal and Prasad, 2000; Huisman and livari, 2002; Karahanna et al., 1999; Mangalaraj et al., 2009). The significance of the difference between factors that impact initial and continued or sustained usage has also been highlighted (Karahanna et al., 1999; Wang et al., 2012). In line with this stream of research, Senapathi and Srinivasan (2012) proposed a model which is an initial formulation of the factors that impact sustained agile usage. The choice of sustained agile usage as a surrogate for success of AMs is consistent with innovation diffusion research (Agarwal and Prasad, 1997; Huisman and livari, 2002). The initial model identifies four categories of factors that influence sustained agile usage: Innovation, Sociological, Technological, and Organizational factors. Within these four categories are nine sustainability factors (two innovation, three sociological, two technological, and two organizational). The four categories of factors on the left hand side are adapted from different theoretical frameworks: agile innovation factors (*relative advantage, compatibility*) are from the innovation diffusion literature (Rogers, 2003), both the sociological factors (*attitude, experience, technical competence*) and the technological factors (*agile practices, tool support*) are from the XP evaluation framework (Williams et al., 2004), and the organizational factors (*top management support, methodology champion*) are from SDM and agile literatures (livari, 1996; livari and Huisman, 2007). A brief review of the initial formulation of the factors is presented below, and Table 1 briefly summarizes the factors in the initial SAUM.

3.1.1. Innovation

Innovation factors relate to individuals' perceptions about innovation characteristics of an innovation. While *relative advantage, complexity, and compatibility* are identified as antecedents to the usage of IS/IT innovations throughout the diffusion literature in general (Agarwal and Prasad, 2000; Mustonen-Ollila and Lyytinen, 2003), only *relative advantage* and *compatibility* have found stronger significance in the acceptance of SE methodology process innovations in particular, including AMs (Hardgrave et al., 2003; Huisman and livari, 2002). This is because significant results in product innovation do not imply significance in SDMs and vice versa, as "the behavioural domain changes from tool use to methodology use, there is a reduction in the relevance of how easy or hard the behaviour is to perform..." (Riemenschneider et al., 2002, p. 1141) which implies that *complexity* does not have high influence on methodology use. Moreover, AMs are aimed at cutting through complexity (Schwaber and Beedle, 2002) and increased agility leads to developers' perceptions that AMs are less complex (Bonner et al., 2010). For example, (Mangalaraj et al., 2009) found strong support for the influence of *relative advantage* and *compatibility* on the acceptance of XP process innovation. A study of developer perceptions in Scrum projects found significant support for the influence of both *relative advantage* and *compatibility*, but ambivalent results regarding perceived *complexity* (Overhage and Schlauderer, 2012a). More recent studies also support this view. A survey of 146 agile practitioners found Kanban practices and tasks to be less complex, did not require a great deal of mental effort to learn, and were easy to use (Ahmad et al., 2016). While complexity may have some impact on initial adoption of AMs (i.e. change from traditional SDMs to AMs) (Chan and Thong, 2009), it may not strongly influence later stages of agile use. Consequently, only *relative advantage* and *compatibility* were included in the Innovation category.

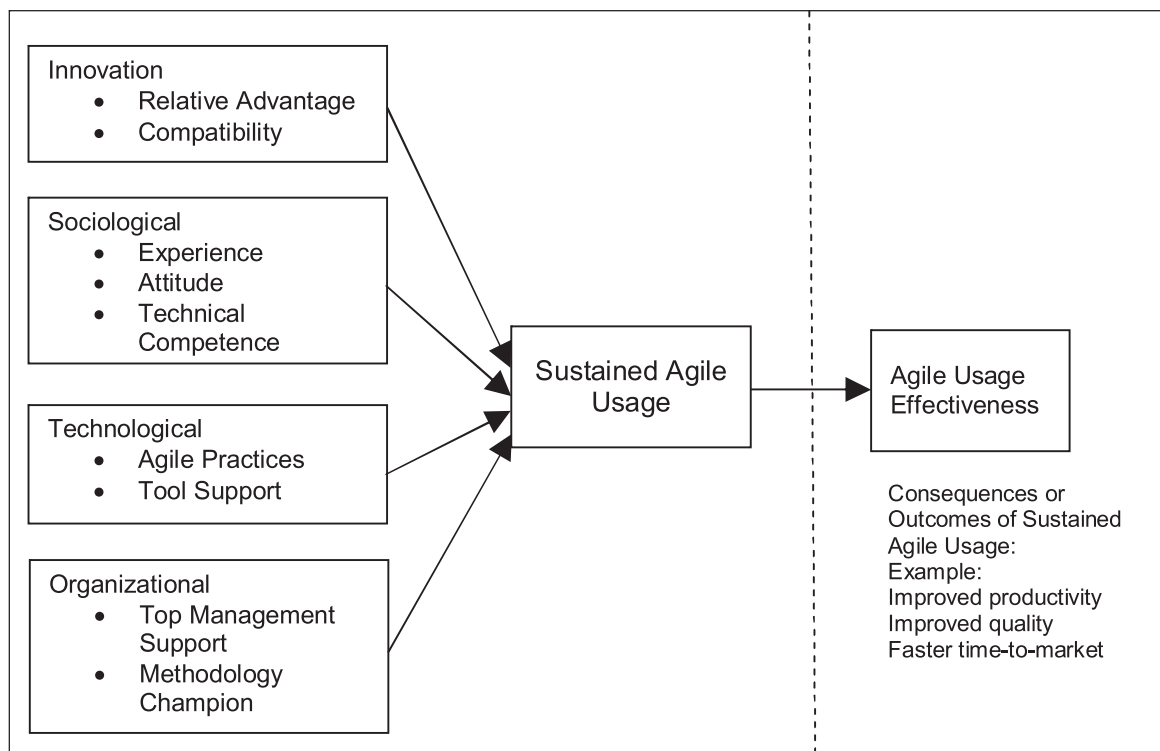


Fig. 1. Initial post-adoptive agile usage model (Senapathi and Srinivasan, 2012).

Table 1

Sustainability factors that influence SAUM (Senapathi and Srinivasan, 2012).

Innovation	
Relative advantage	The degree to which an agile method is perceived as better than its precursor methods.
Compatibility	The degree to which agile methods are perceived as being consistent with the existing practices, values, and past experiences.
Sociological	
Experience	The number of years of working experience in systems development.
Attitude	The prominent positive or negative beliefs about the continued use of agile methods.
Technical competence	The technical knowledge, expertise, and domain expertise.
Technological	
Agile practices	The use of engineering practices, such as continuous integration and refactoring.
Tool support	The incorporation of tools to support effective use of engineering practices.
Organizational	
Top management support	A balanced approach where management provides ongoing support at appropriate points through all stages of assimilation.
Methodology champion	A championship role instrumental in the effective assimilation of agile methods throughout the organization.

3.1.2. Sociological

Sociological attributes will influence later stages rather than earlier stages of an innovation's assimilation (Gallivan, 2001). In particular, three factors (*experience*, *attitude*, and *technical competence*) have been found to have a significant influence on the long-term and effective use of both SDMs and AMs (Agarwal and Prasad, 2000; Williams et al., 2004). For example, high levels of *experience* and *technical expertise* have been reported to have a significant influence on the spread of use of XP to a large number of teams (e.g. over 30 teams) (Layman et al., 2004; Williams et al., 2004), and also on the sustained use of popular AMs in medium to large scale organizations (Russo et al., 2013; Senapathi and Srinivasan, 2012).

3.1.3. Technological

While agile values bring purpose to *agile practices*, *agile practices* are evidence of values, which are clearer, more specific, and expressed on a higher level (Abrantes and Travassos, 2013). Practices bring accountability to values and are activities that implement the principles governing agile methods (Abrantes and Travassos, 2013). Therefore, it is logical that the effective use of *agile practices* are critical enablers of sustained agile usage (Senapathi and Srinivasan, 2012).

Examples of widely used *agile practices* are test driven development, continuous integration, planning game, on site customer, collective code ownership, metaphor, and refactoring (Abrantes and Travassos, 2013).

While *agile practices* pertain to technical practices such as test driven development, and continuous integration, *tool support* refers to the effective implementation of agile practices (e.g. Cucumber for automated testing, CruiseControl for test driven development) using the appropriate tools (Williams et al., 2004). It has been demonstrated that the choice and effective use of the right combination of agile practices and tools has significant benefits on agile usage and improvements in development outcomes (Abrantes and Travassos, 2013).

The influence of technological factors on the initial adoption of agile use is well supported in the literature. For example, (i) Chow and Cao (2008) found significant support for the impact of *agile software engineering techniques* (e.g. well-defined coding standards, simple design, refactoring) and (ii) Misra et al. (2009) found significant relationships between sociological characteristics and the success of agile projects. Their positive influence on the long-

term or sustained use of AMs has also been recognized (Mangalaraj et al., 2009; Russo et al., 2013; Senapathi and Srinivasan, 2012).

3.1.4. Organizational

Organizational factors include *top management support* and *methodology champion*. Sustained and effective usage of AMs represents a long-lasting and substantial departure from traditional systems development approaches and calls for careful change management. This requires strong commitment, consistent involvement, and support from the top management. *Top management support* is most consistently reported to facilitate any IS implementation initiative. In particular, the significance of its influence on all stages of a systems development methodology's assimilation (e.g. both initial stages (adoption) and later stages (acceptance and usage)) is well established in the systems development methodology literature, including AMs (Huisman and Iivari, 2002; Vijayasarathy and Turk, 2008).

Methodology champions are characterized as bearers of persuasive and evaluative information about an innovation who not only promote the users' vision of using an innovation but also use a variety of other influences (e.g. social, political) to eliminate barriers to successful diffusion and implementation (Beath, 1991). In the context of AMs, *methodology champions* play a significant role in facilitating the ongoing usage of agile practices, which include responsibilities such as convincing management, pushing through implementation hurdles, and mentoring (Russo et al., 2013; Senapathi and Srinivasan, 2012; Wang et al., 2012).

3.2. Agile usage effectiveness

On the right-hand side of Fig. 1, *Agile Usage Effectiveness* refers to consequences or improvements in outcomes resulting from sustained agile usage (Senapathi and Srinivasan, 2012). It is based on the reasoning that if agile methods are not sustained effectively throughout the organization, we are not able to ascertain their impact on systems development activities. The list of possible outcomes shown in Fig. 1 is not exhaustive and may be perceived differently by different stakeholders. For example, outcomes to the agile development teams may include improvements in productivity and quality of the development process and predictability. On the other hand, outcomes visible to the business division may include customer satisfaction and faster time-to-market. The vertical dotted lines are used to indicate that our focus is on the left-side hand of Fig. 1.

4. Research method

A focus group comprised of 29 practitioners from 21 different organizations, followed by 20 qualitative interviews with practitioners from five organizations separate from those in the focus group were employed in this research. Interviews lend themselves well to be used in combination with other methods in a multi-method approach as adopted in the current study (Robson, 2011). Focus groups are well suited for the initial phases of research to help prepare for the main data collection phase, for example as a precursor to the development of a more structured instrument (Robson, 2011). In this study, the focus group was conducted before the interviews. It was used to examine emerging ideas, including collective and individual perspectives, and meanings in which various practitioners perceived, experienced and understood the various concepts that influence sustained agile usage (Ritchie et al., 2013). Interviews were then used to further refine the findings of the focus group and explore in more detail the perspectives and opinions of expert practitioners on the critical factors that impact sustained agile usage.

4.1. Phase 1: focus group

Focus groups aid in collecting information from members of a clearly defined target audience on a clearly defined topic (Morgan, 1997) such as sustained agile usage. Focus groups not only collect rich data from a varied group in a short amount of time, but they allow participants to probe other participants' reasons for their viewpoints, even challenging other's viewpoints (Morgan, 1997; Stewart et al., 2007). The group interaction aspect is particularly relevant as it allows insights and important data to emerge which might not otherwise surface (Kidd & Parshall, 2000; Kitzinger, 1997). In the current study, the focus group provided an opportunity to explore the varying viewpoints of practitioners who worked on different agile systems development teams to refine and expand the SAUM model to include additional factors that either facilitate or hinder sustained agile usage.

The focus group consisted of 29 participants who divided themselves into five smaller focus groups of 5 to 6 participants to hold detailed discussions on each of the SAUM categories and their associated factors. The 29 participants were software industry agile practitioners, which is a relatively large focus group as they usually consist of 6 to 12 participants (Johnson and Christensen, 2004). But larger focus groups are recommended for topics where researchers want to collect multiple perspectives and opinions (Morgan, 1997). Thus, we used a larger focus group because our research objective was to refine the SAUM by seeking multiple perspectives. We divided this larger group into the smaller groups to discuss their factors. Then, we brought the smaller groups back to one larger group of all 29 participants for a discussion on the findings in their small groups.

Participants were attendees at a professional software development conference with 1700+ attendees. It was hosted in the USA, and although it was an international conference, all participants in this focus group came from the USA. They self-selected to attend this group with a choice of 19 other sessions offered at the same time. While this is a limitation of the focus group, it does provide participants focused on learning how to sustain the use of agile methods in their teams and organizations and thereby a viable group to discuss the SAUM.

Table 2 shows the participants had a range of experience from beginners with agile methods to those working with agile for 15 years, with the average experience with agile development being 4.14 years and the median experience being 4.0 years (see Table 2). Six participants had less than 2 years of experience. Thus, we include a range of experience from participants, and some data was provided by those fairly new to agile methods compared to those with many years of experience. While we could criticize the use of less experienced agile members as inappropriate to a study on agile sustainability, we argue instead that these participants ensure heterogeneity in terms of experience, leading to a more dynamic dialogue.

Additionally, participants had diverse backgrounds with varying industry sector experience with agile development. Participants listed the AMs they use as Scrum, XP, Kanban, Lean or a combination of these four methods. Their roles included agile practitioners, agile coaches, product owners, and business analysts. But the participants were not asked to differentiate themselves based on role or organize by role. The larger group was divided into five smaller groups by allowing participants to randomly create their own groups. This process facilitated a quick, random grouping as the nature of the venue did not provide time in advance to create the groups because we did not know who would attend the session until we were facilitating the session. It is a slight limitation of this data set that the researchers did not create the groups. The researchers assigned one seed category, i.e. one of the four sustainability categories from the initial SAUM, to each of these five

Table 2
Profile of focus group participants.

Participant Criteria	Participant Response
Number of Participants	29
Mean Experience with Agile Development	4.14 years
Median Experience with Agile Development	4.00 years
Least Experience with Agile Development	0.25 years
Most Experience with Agile Development	15.00 years
Mean Size of Organization	19,950 employees
Smallest Size of Organization	1 employee
Largest Size of Organization	100,000 employees
Mean Number of IT/Systems Development People in Organization	1542 employees
Smallest Number of IT/Systems Development People in Organization	1 employees
Largest Number of IT/Systems Development People in Organization	16,000 employees
Role	Developer (6) Scrum Master/Project Manager (13) Quality Assurance Tester (3) Agile Coach (3) Business Analyst/Director (4)
Industry Segments	Consulting Defense Financial Services (e.g. Banking, Insurance) Government Health Care IT Manufacturing Transportation Travel

smaller groups for that group to discuss. While this could also be viewed as a slight limitation to this research, we wanted to ensure we collected data for all four categories rather than risk all small groups discussing only one or two of the categories. By assigning teams to a category, we ensured we had data for each category. These smaller group discussions were then shared with the larger group so that others could comment on the data.

The duration of the focus group was 90 minutes. The structure of the focus group was an exercise and open discussion on sustained agile usage via horizontal and vertical usage. First, the researchers gave a general overview of the five stages of how innovations diffuse in an organization: adoption, adaptation, acceptance, routinization and infusion to explain why they were developing an SAUM. Then they gave a brief overview of the components of the SAUM to define what was meant by sustainability factors, horizontal and vertical usage, and agile usage effectiveness. These were described to set the stage for the groups' discussions and did not lead the participants in any one direction to define factors.

After this brief presentation of the above-mentioned topics, each of the five smaller groups conducted the activity to apply the SAUM. Each group member offered examples from their own agile teams that illustrated the sustainability category assigned to their group. They discussed examples that either enhanced or hindered the sustainability of agile methods. The researchers had previously prepared worksheets for each group to write down their examples that enhanced and their examples that hindered the sustainability of AMs. Participants were instructed to select and present as the strongest example that which their group felt enhanced or hindered the use of agile methods the most.

After the participants had completed this activity, the researchers facilitated the discussion on each group's strongest example for enhancing and for hindering the sustainability of agile methods. If their examples did not fit their allocated category or they had discussed a new, more appropriate factor, they presented this finding as well. Thus, we conducted the focus group in a responsive manner (Rubin and Rubin, 2005; Wengraf, 2001) to follow up on insights uncovered mid-session and adjust the content and schedule of the focus group accordingly. All 5 groups participated in this discussion. Although it was a large group, this re-

sulted in a lively and engaging discussion. The entire session was audio recorded.

4.2. Phase 2: Semi-structured interviews

The focus group provided a range of perspectives and diverse opinions as it included participants with varying levels of agile experience, with varying agile roles, from various industries, thereby creating a generalizable group. It served as an initial phase to help prepare for the interview phase. It was a precursor to the development of a more structured and refined data collection instrument for use during the 20 face-to-face semi-structured interviews. These interviews enabled further refinement of the focus group findings and provided additional insights on sustained agile usage.

In the IS discipline, semi-structured interviews are considered appropriate where research and theory are in their early formative stages (Benbasat et al., 1987). They allow investigation of an IS phenomenon such as sustained agile usage as perceived by its practitioners and users and enable research in which few previous studies exist. Semi-structured interviews' main characteristics of individual responses as the source of data, the researcher's main role in data collection, and a focus on stakeholder perspectives and perceptions (Dubé and Paré, 2003) matched the requirements of the current study.

The interview design consisted of 5 organizations and 20 interviews. Interview participants were separate to the focus group attendees and were mainly from New Zealand. The participants came from a range of organizations and a wide range of roles, such as developers, business analysts, project managers (PM), scrum masters (SM), and IT service delivery managers, including stakeholders from the business unit and management. All participants had a minimum of 2 years' experience of working with agile methods. The organizations ranged from consulting and professional service providers (Org A), banking and retail (Org B), governmental bodies (Org C and E), and insurance providers (Org D). Tables 3 and 4 show the profile of the interviewees and the roles respectively.

All 20 interviews were conducted in person by one researcher. Interviews ranged from 60 to 90 minutes. The interview protocol guide approach was used as it is: (i) more comprehensive and systematic for data collection than the purely conversational in-

Table 3
Profile of interview participants* [*Participant N = 20].

Experience	Years
Mean Experience with Agile Development	5.5 years
Least Experience with Agile Development	2 years
Most Experience with Agile Development	10 years

Table 4
Roles and industry sector.

Industry Sector	Roles Interviewed
Professional Service providers (Org A)	Agile Expert/Coach (2)
Banking and Retail (Org B)	Business Analyst (1)
Government Organization (Org C)	Developers (4)
	Tester (1)
	Business Analyst (2)
	Scrum Master (2)
Insurance (Org D)	Developers (2)
	Business Analyst (2)
Government Organization (Org E)	IT Delivery Manager (1)
	CIO (1)
	Product owner (1)
	Project Manager (1)

interview, and (ii) more flexible than the standardized open-ended interview or the closed, fixed-response interview (Agerfalk and Fitzgerald, 2008).

Interviews were recorded, and subsequently transcribed, proof-read, and annotated by the researchers. Field notes and analytical memos were created. Any issues that emerged following the interview transcription process were clarified using either informal follow up interviews or via email within the subsequent few days.

5. Data collection and analysis

For phase 1, a focus group protocol was prepared based on the initial SAUM. The four sustainability categories of the SAUM provided a list of intellectual bins, or seed categories, (Miles and Huberman, 1994) to structure the data collection and the open coding stage of data analysis. A similar approach was followed by one of the very few focus group studies in IS where an initial conceptual framework was used as a lens to guide the discussion and structure data collection and analysis (O'hEocha et al., 2010); thus, there is precedence for how we conducted our focus group. In this study, we used four intellectual bins to represent each of the four sustainability categories of the model: Innovation (2 teams), Sociological (1 team), Technological (1 team), and Organizational (1 team) categories. The participants were divided into 5 teams with each team assigned to 1 seed category, i.e. 1 of the 4 sustainability categories. Two teams were assigned to the Innovation category because there were enough participants for an additional team, and we focused on agile use as an innovation in the organization. We assigned teams to each category to avoid teams talking only about 1 or 2 of the categories, leaving us with no further data for model refinement. By assigning teams to a category, we ensured we had data for each category. The goal was for each team to discuss examples from their own agile teams/organizations that illustrate each category assigned to their group. If their examples did not fit their category or they had discussed a new, more appropriate factor, they presented this finding as well. While one researcher facilitated the session, another listened, observed, and took notes. The researchers then switched roles to account for any variance between their note-taking and questioning. In any cases of ambiguity, clarification was sought from the attendees during the focus group.

In order to aid analysis, the written worksheets we previously discussed where participants wrote out their examples of what hindered or enhanced their use of AMs were transcribed and the focus group was both recorded and transcribed. Transcriptions were then proof-read, annotated, and coded by the researchers using NVivo software, a qualitative data analysis software package. The data (i.e. sustainability factors with examples of those that enhanced and hindered agile sustainability) were also emailed to all participants for feedback and validation. No participants sent edits or changes to the data, except to comment on the usefulness of having the data for their work at their own organizations.

For phase 2 with the interviews, we developed an interview protocol to guide the data collection. Dialogical reasoning requires sensitivity to possible contradictions between the theoretical preconceptions guiding the research design and actual findings (Klein and Myers, 1999). While the factors identified in the model served as a conceptual starting point, participants were given the opportunity to openly discuss any additional factors or measures relevant to sustained usage. For this reason, they were asked about any key projects, key milestones, resources, as well as any critical challenges that they perceived to have significant impact on sustained agile usage. Interviews were transcribed and then coded based on the seed categories represented by the four categories in the initial model as was similarly done with the focus group data.

As participants tried to relate their input to specific concepts, newer themes that emerged were classified into new nodes, which is the term to indicate a specific concept or theme in NVivo. As input was initially focused on the discrete set of sustainability factors in the initial model, differing understandings and ambiguities about these factors also emerged. After the first few interviews, the terms used to describe some factors were modified, for example, “*experience*” was modified to “*agile experience*”. This supported a more coherent understanding and interpretation across participants. The data relating to each factor were assessed in relation to the category they belonged to and whether they facilitated or hindered sustained agile usage.

Next, we developed analytical memos as new themes or factors emerged from the field notes. We combined this with an analysis of the transcripts from the focus group and allowed new or refined factors to emerge. We used the constant comparison method (Gallivan, 2001; Patton, 2002; Strauss and Corbin, 2007) to compare focus group and interview data sets, i.e. to check whether the categories and factors from the initial model and the focus group data still applied to the interview data or whether any modifications were required. The initial set of nodes was refined as the analysis evolved, and new nodes were created for any potential new factors that were identified during the process. The coded factors were re-analyzed to ensure that they belonged to the correct category. We continued doing this until we were confident that the categorization we developed was well supported by the data. Fig. 2 outlines our process for analyzing the various methods of data collection.

In qualitative research, the problem of multiple realities refers to the fact that the understanding of reality is based on individual interpretation of data and that different individuals may interpret the same data in different ways (Kaplan and Duchon, 1988). We addressed this problem in two ways: first, our analysis approach uses conscientious coding which enabled us to create a traceable audit trail of the research process from data collection through to the drawing of conclusions. This also helped in improving the reliability and repeatability of the research. Second, we vetted the data, whereby results and interpretations were discussed with professional colleagues. The findings were also presented and discussed with colleagues in reputed practitioner/academic conferences and workshops.

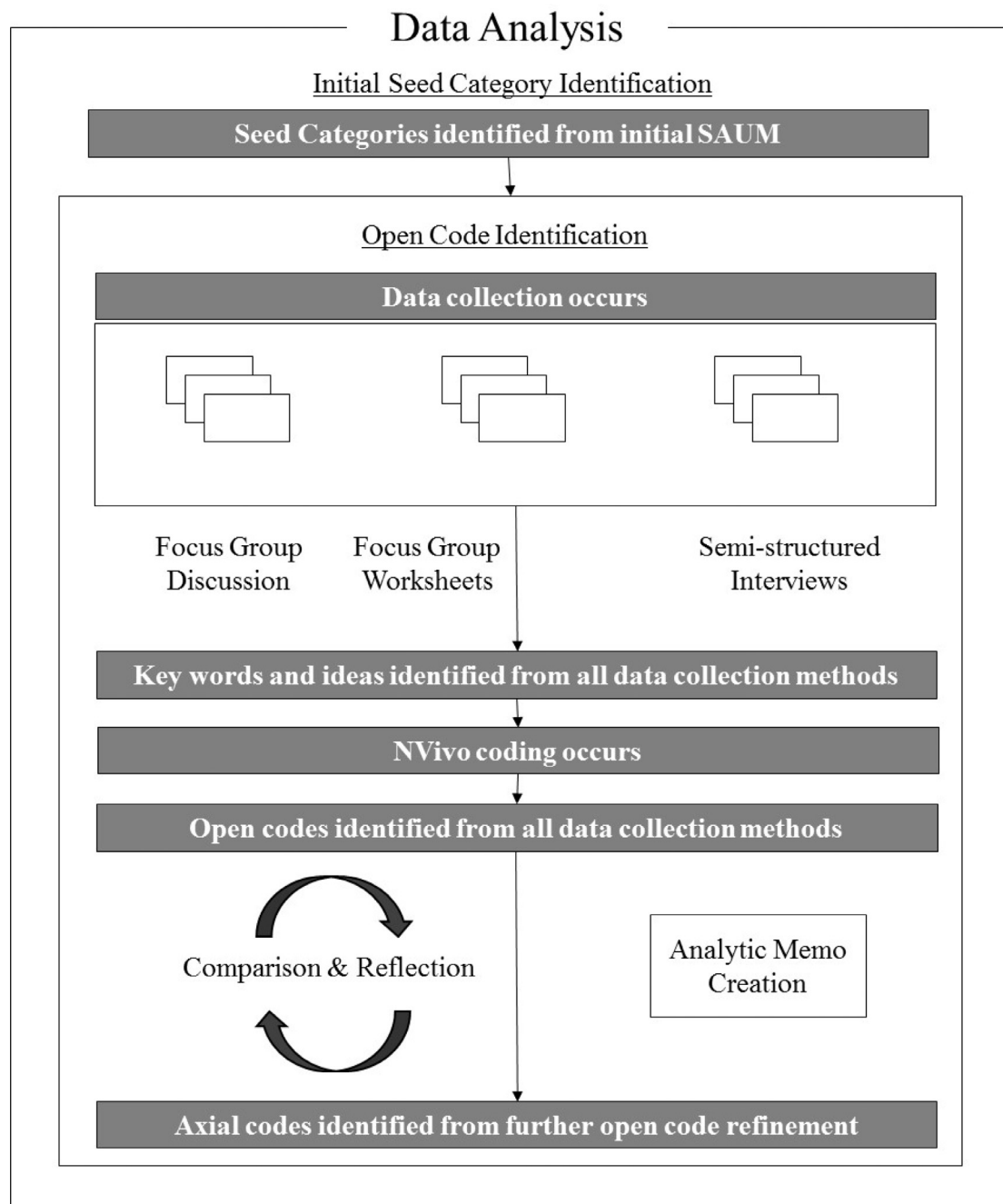


Fig. 2. Process used for analyzing all methods of data collection.

6. Findings and discussion

In the following sections, we discuss the integrated findings from both the focus group and the interviews. We present the refined set of factors under each of the four categories of the initial SAUM.

6.1. Innovation [removed from the refined model]

Two constructs constitute this category: *relative advantage* and *compatibility*. While there was a general consensus on agile methods being perceived as better than the precursor methods from both interview (5 citations) and focus group participants, evidence of tracking improvements in the agile development process was

found to be more important for sustained usage of AMs. From our data, 70% (13 specific citations) of interview participants discussed that tracking appropriate metrics was critical for the continued use of AMs. Participants explained how tracking project release progress using burndown charts, velocity to track the development process, and release burndowns (relative sizing to track velocity over a number of sprints projecting project completion time) were critical in measuring improvements in the development process. The significance of metrics from clients' perspectives was also highlighted:

Resource burn rate is a burn-up chart that is used to manage and forecast project spend based on resources allocated to the project. This metric can be very important for the clients too as it helps

them to manage their budgets and when used in conjunction with the release burn can form the basis of Earned Value Analysis (EVA). [Developer, Org D]

Tracking such metrics was deemed critical to sustain the efforts of initial adoption initiatives. This is well expressed in the words of a Scrum Master at Org C:

It's like a little bright flame like this scrum which is working really well and it sort of extinguishes itself without such evidence. We are going to degrade into what we've done previously – it's a sustainability question.

Participants also stressed the importance of tracking metrics from business perspectives (i.e. measuring improvements in customer satisfaction, complying with market need and expectations) in terms of tangible business benefits, as an IT Service Delivery manager at Org E explained:

Providing metrics to report back to the business to say that we are doing better. I guess on measuring success, perhaps measurements on improvements in business value, things like, 'Am I able to process my survey in half the time than I did before? Is my data better than it was before?'

Focus group participants also expressed similar views, i.e. they highlighted the significance of facilitating and “...promoting successful agile experiences to others in the organization through customer feedback surveys, retrospectives, and agile forums” [focus group participant].

In summary, tracking both agile (measuring aspects of the development process) and business (measuring aspects of improvements in business value) metrics that demonstrate the team's progress throughout the development cycle was found to have a significant influence on sustained usage. For example, by tracking and monitoring how velocity evolves over time, other teams can expect to see an increase in velocity as the team's development process is optimized. The findings indicate that evidence of metrics that demonstrate improvements in the development process was more important. In other words, while the perceptions on the benefits of using agile may facilitate initial stages of adoption, making the results or benefits apparent to others facilitates later stages of continued or sustained use. This theme that emerged from the analysis of both interview and focus group data was found to relate more closely to the *result demonstrability* construct, which is the degree to which the results of an innovation are visible to others (Rogers, 2003) rather than to the *relative advantage* factor used in the initial model. Therefore, *relative advantage* was replaced with *result demonstrability*. Moreover, as agile metrics are aimed at measuring the team's performance and providing measurable goals for the team, it was deemed more appropriate for *result demonstrability* to be part of a Team Factors category rather than the initial Innovation category.

Analysis of both focus group and interview data on *compatibility* can be classified into two main themes: (i) AMs being consistent with existing values and practices (3 specific interview citations) and (ii) incompatibility in funding and resourcing models, and inability to leverage enough management support (4 specific interview citations). Findings in support of the first theme indicated that while there may be minor compatibility issues related to the adaptability of agile practices during the initial adoption period, they do not have significant influence on the later stages of an innovation's use, i.e. their continued use. Findings in support of the second theme highlighted compatibility issues from an organizational perspective. In particular, the significance of having an organization structure that is compatible with agile methods was stressed. The significance of addressing such compatibility issues

by supplementing agile use from a governance perspective was highlighted:

You need to supplement it from a governance perspective. By itself it is not going to succeed – so I guess really compatibility is more of integrating it throughout the organization. Otherwise, another project comes along, and so people revert into previous types, and so a lot of compatibility is around acceptance at the governance level, basically at a higher management level as opposed to just delivering as a project...because they might require dedicated resources, teams feel empowered, negotiate on scope, dedicate time for testing – so a lot of this [is] around the compatibility of the organization to supporting agile. [Project Manager, Org E]

From these findings, it was deemed more appropriate to add a factor called *organization structure* under the Organizational category, and to remove compatibility from the Innovation category.

Some interview participants (3 citations) also shared their perceptions on whether AMs were difficult to understand and use. These findings provided some insights on the *complexity* factor, which was not included in the initial model. For example, a Business Analyst at Organization D explained:

Initially yes, it's a new way of doing things, and you have to learn how to do it. Initially a few sprints were needed to adjust to it, but there were few issues, for example delays in decision making, too many tasks in one sprint. But you learn as you go through. You attempt too much. And so as a new team you don't know your velocity, so you try to attempt too much and then realise, 'no it's not right'. And we had difficulties early on like user stories- not breaking them up, and not having them at the right level, so we had to break them up. So we learn as we go through.

And a developer at Org C stated that learning happens “...as you go, all the practices such as daily stand-ups, really the steps are fairly well known; all the rituals and ceremonies and all that sort of thing are quite well known”. The findings suggest that *complexity* may not have significant influence on the sustained use of AMs. While this is consistent with most studies in the process innovation literature (Ahmad et al., 2016; Hardgrave et al., 2003; Mangalaraj et al., 2009), other studies have found ambivalent support for the influence of complexity on long-term usage of AMs (Overhage and Schlauderer, 2012a).

However, it should be noted that *perceived complexity* in Overhage and Schlauderer (2012a)'s study is conceptualized using two factors: *task complexity* and *discipline*. Out of these, only *task complexity* aligns well with the definition in the innovation diffusion literature, i.e. it taps the aspect of whether the underlying ideas and practices of agile are easy to understand or apply. Their ambivalent findings (i.e. higher perceived complexity) relate more to the specific challenges associated with the required *discipline* rather than on individual *task complexity*. They found that *task complexity* aspects such as smaller development increments and individual tasks were less complex and easier to manage, whereas *discipline* aspects such as up-front planning in starting new projects and management of external dependencies in the execution of large distributed projects were more complex. This implies that complexity may have stronger influence on the discipline aspects of large distributed projects rather than on the individual tasks and practices related to the usage of AMs. This is consistent with the initial justification on which this study's initial model was developed.

In summary, *relative advantage* was changed to *result demonstrability*. Furthermore, given that agile metrics are only relevant and applicable at the team level, *result demonstrability* was moved to the Team Factors category, which was the Sociological category in the initial model. With no factors left in the model, we removed the Innovation category from the model.

6.2. Agile team factors [the sociological category in the initial model]

The initial Sociological category had three individual factors: (i) *experience*, (ii) *attitude*, and (iii) *technical competence* that were perceived to make sustained usage more or less likely to occur. Both interview and focus group data provided new insights on these attributes and led to changes in our model.

6.2.1. Agile experience [changed from experience in the initial model]

While experience was identified as an important factor, a high level of experience in systems development by itself did not appear to be directly related to sustained agile usage in many participants' organizations. While some interview participants (2 specific citations) perceived a high level of systems development experience might drive against agile use, others (2 specific citations) suggested that high levels of experience may provide the confidence to use new methods and practices. A combined view of such perceptions is well expressed in the following statement by an interviewee:

The more experienced people are, people have more confidence and so sometimes they can be open to try something new. On the other hand, it might be that if they have experience, then they don't want to change, and in that way experience might work against agile use. [Developer, Org D]

Several participants perceived that more experience in a role with systems/software development or that domain may not always have a positive impact on sustained agile usage. Lack of specific experience in AMs may actually hinder the effective use of agile methods in organizations. An interviewee explained how employees who had been working for their entire careers in their organization had gained expertise in both technical and business domains, but their lack of experience in agile methods had a negative influence on sustained agile usage. A lead developer from organization C explained how losing experienced agile staff for newer members with no agile experience hindered sustained agile usage in her organization:

We have new project managers coming on that don't know agile. You've got testers coming in, BAs that have never done agile.... we had a really good [agile] experience; we did it really, really well. And now all those people are gone and now we're just left with a couple that are on the fringes or those that are less dynamic. So now we're having a real [difficult time]... and to make it sustainable you sort of need these people who are living and breathing it, i.e. you know you've got to go through it to see the level of commitment and how to plan it all.

The above findings not only stress the significance of *agile experience* (9 specific citations) on sustained usage, but also highlight the need for a clearer differentiation between *experience* and *agile experience*. *Experience* in general refers to a person's length of experience in a role or an organization. It is associated with an individual's prior technical knowledge and expertise (Brooks, 1980). Therefore, it is logical to expect that more experienced people will have a greater ability to recognize the benefits of innovations such as agile and therefore more likely to positively influence its adoption (Chan and Thong, 2009). While such experience may have a positive influence on the initial stages of agile adoption, findings from this study highlight that specific experience in using agile methods (i.e. *agile experience*) rather than general experience is critical in facilitating the later stages of use as it influences practitioners' willingness and ability to adapt to changes in values and principles associated with agile. For example, Salo and Abrahamsson (2008) found that people who had specific experience with AMs such as XP and Scrum had more favorable opinions about their usefulness and discovered a positive association

between length of agile experience and attitudes towards its usefulness (Laanti et al., 2010).

The need for *agile experience* to be manifested at the team level was also consistently cited by both interview and focus group participants. Relatedly, Manen and Vliet (2014) report a number of positive benefits of dedicated teams (with team experience), such as increased trust and collaboration, which were found to facilitate the expansion or scaling AMs beyond their initial use. It is reasonable to expect that high degrees of in-house team experience in AMs should enhance the propensity of continued agile usage. Consequently, the factor *experience* was renamed *agile experience*.

6.2.2. Agile mindset [added to the refined model]

Prior research on the influence of sociological attributes such as *attitude* and *motivation* on the adoption and use of IS innovations has mainly focused on individuals as the adopting unit (Agarwal and Prasad, 2000; Chan and Thong, 2009; Mangalaraj et al., 2009). However, findings from this study provide new viewpoints: participants identified and stressed the significance of a number of important characteristics to exist at the group or team level and organizational level rather than at the individual level. Characteristics such as resilience, commitment, willingness to fail, intrinsic motivation, and ability to improve, learn, and adapt, were identified as key influencers of sustained agile usage. Phrases such as "agile mindset", "inclusiveness", "unselfishness", "agile mentality", "emotional intelligence awareness", and "team spirit" were frequently used to describe these concepts.

For example, seeing daily stand-ups as a safe-circle for improving their sharing about what they have done with their team rather than reverting to just sharing their work or story required an attitude of:

Does anyone else need a hand now, can I help you with this or do your filing?" That does not come naturally, so you need that inclusiveness, its unselfishness and it's that emotional awareness isn't it, about what everyone else is doing. So, you need people with the right attitude. [Developer, Org D]

The focus group highlighted "chemistry of the team" as the strongest sociological factor that facilitates sustained agile usage:

We discussed a particular team that had everything going against them, they had a legacy, they had different vendors, they had on-shore, off-shore teams. But the team developed the chemistry, i.e. the communication between the teams, ownership of whatever they were working on, and also the leadership. So, the chemistry of the team was our big example of a sociological factor.

Attitude by itself generally refers to an individual person's expression of favor or disfavor towards the use of an innovation based on his or her individual beliefs, preferences, and intentions (Agarwal and Prasad, 2000). While such attributes (e.g. attitude, motivation) are commonly associated with an individual mindset, the concept of the group/organizational mindset has also been introduced to distinguish between automatic, mindless and conscious, mindful information processing (Walsh and Ungson, 1991). This relates to the concept of collective mindfulness, characterised by "self-regulated", "continuous attention to detail", "vigilance to minimise errors and respond effectively to unexpected events" with a strong emphasis on "team focus" and "team commitment" rather than "individual focus" or "individual mindset" (McAvoy et al., 2012; Vidgen and Wang, 2009). In AMs, this is commonly referred to as the *agile mindset* which encapsulates a number of characteristics as discussed above, i.e. agile values guided by agile principles, manifested through many different practices such as Scrum or XP (Sidky, 2012).

Relatedly, Manen and Vliet (2014) identify a number of *agile mindset* characteristics' (e.g. team thinking, taking responsibility,

willingness to try new way of working) that facilitate agile expansion in large organizations. This concept reinforces the findings from this study which highlights the need for the *agile mindset* to become so established within the teams that they intentionally continue the associated behaviors to the extent that it becomes a norm or way of life. This was recognized as key to sustainable agile transformation. Without such a mentality, resistance builds up resulting in dysfunction within teams and people revert back to their old traits, especially under pressurized conditions. Therefore, *agile mindset* was identified as a sustainability factor because of the strong support it received both from the focus group and interview data (11 specific interview citations).

6.2.3. Technical competence [removed from the refined model]

There were five specific citations that related to *technical competence* within the interview data. Out of these, two citations overlapped with the *agile experience* theme, and three with the *agile mindset* theme. While *agile experience* citations captured elements related to specific knowledge and expertise in AMs, more general technical skills and expertise such as programming, .NET, and SQL skills were identified as characteristics of high performing teams with an *agile mindset*. The importance of these attributes at the team level was consistently emphasized, meaning that the lack of individual technical competence and expertise may not have a significant influence on sustained agile usage as long as these attributes are evident at the overall team level. A business analyst at Org D explained:

So the technical expertise and knowledge is not of one person and the other person can sort of support it, so when there was one less knowledgeable we supported that person to pick up, and we allowed for that knowledge gap when we did the estimates for the team, for this person to pick it up, i.e., we put the estimate up, and decided to break things down, so we sort of accommodated for that. So, the overall team competence could still be rated as high.

The findings also stressed the importance of forming well-balanced teams:

So you need a good balance and variety where people can learn from one another. So you need to design in that variety in your teams in order to get a good result in the long run, and such teams could lead to lots of learning teams that grow very rapidly because those high performance teams can even teach other teams, everybody. [Agile coach, Org A]

Such teams were believed to strive to continually push the teams to cope with any extra challenges. As discussed above, 5 citations overlapped with the factors *agile experience* and *agile mindset*, but there were no specific citations that could be attributed to *technical competence*. Given this weak evidence, *technical competence* was removed from the model.

6.2.4. Agile coach [added to the refined model]

A new factor, *agile coach*, was raised and cited in both focus group and interview data. It was identified as a key factor in creating and sustaining well-balanced high performance teams. It received strong support from focus group participants and 12 specific citations in the interview data. A coach was identified as someone who possessed both technical and business expertise and had the ability to both challenge teams' perceptions of what they're capable of and allow them to find their self-organizing behavior. An *agile coach* is normally someone external to the organization who temporarily becomes part of the team or someone internal who has the energy and enthusiasm to actually take the team forward. Comparing software intensive systems development to a high performance sport, an agile coach at Org A explained:

If I were to say, 'What's the number one factor in sustaining continuous improvement?' I would say, 'Are you treating your organization as a high performance organization, are you coaching your organization?' If you aren't, then coach that organization to sustain that high performance...".

This perception was also echoed in the focus group, "*The best way that we know of right now to create and maintain high performance teams is to have a coach*" .

An *agile coach* was identified as a role (most often external) associated with skills and expertise in professional (leadership, facilitation, mentoring), technical (agile), and business (application, subject matter) domains. The role involves helping agile teams (*knowing* and *doing* the agile process and practices) in transitioning themselves to motivated, high performance teams. While traditional leadership roles such as Project Manager and Scrum Master are compatible with the agile coaching role, some key differences were also highlighted. Roles such as Project Manager have a wider set of project and organization-specific responsibilities, including reporting progress, performance appraisals, and pressure to deliver. These responsibilities take the focus away from process improvement, whereas coaching is a transitory role whose main focus is on teams becoming self-coaching and adept in applying agile practices. The significance of formal coaching roles on sustaining and improving effective agile use were stressed by many participants. An agile expert at Org A further explained that while investing in specialist roles such as coaching was vital, the critical factor for sustainable usage relies on developing and fostering in-house coaching capabilities and communities:

You have to build the system to make it happen...we need to develop our own coaches ourselves. We need to be able to coach others, right? We need to have a good system for creating good agile coaches, helping them to grow others, and develop a system that creates this self-perpetuating improvement.

In summary, while the positive influence of coaching was recognised (including short-term external coaching), the key to achieving long-term sustainability is to focus on developing and empowering internal coaching communities so that organizations can be self-sufficient in sustaining effective use.

It is clear from the above discussion that sociological attributes are expected to be manifested at the team level. It is indispensable that they are internalized at team levels rather than just implementing agile as a process. It requires that people "*work as part of a team rather than as individuals. [This is] key to sustainability, i.e. moving from an individual culture to a team culture..*" [Agile coach, Org A].

A Scrum Master at Org C captured this clearly:

So I think the whole thing at a sociological level is: is it moving from being an individual and optimizing the outputs of the individual, to optimizing the outputs of the team, the program, or the business unit and then the organization. So for me, I think that is key to wider use, wider acceptance of agile.

Consequently, in light of the above discussion, the Sociological category was re-specified as Agile Team Factors with four factors: (i) *agile experience* (changed from experience), (ii) *agile mindset* (characteristics of attitude and technical competence combined), (iii) *agile coach*, and (iv) *result demonstrability* (changed from relative advantage and moved from the Innovation category).

6.3. Technological [changed to technological factors in the refined model]

The initial model had two factors in this category: *agile practices* and *tool support*. Both interview and focus group data strongly sup-

ported the positive influence of *agile practices* (7 specific interview citations) and *tool support* (13 specific interview citations):

Yes, absolutely. So for example, the ability to do regression testing after it's reached its duration, and the ability to do that quickly and efficiently is critical. We are still quite reliant on manual progression tests which are quite labor intensive and have a breaking effect on success, so that has quite an important factor to play. [Developer, Org C]

The significance of specific practices such as automated testing and refactoring was stressed, “So from a technology perspective, refactoring and ease of automation of testing are probably the two key things” [Agile coach, Org A], and the influence of using appropriate tools to support effective use of *agile practices* (e.g. Junit for automated testing for Java and TeamCity for continuous integration) was reiterated:

What a lot of people don't realise is that we are at war. We are at war with complexity. So being at war with complexity for any non-trivial amount of code, as long as I've written anything substantially more difficult than, 'hello world,' I simply cannot afford to have one line of code without some supporting test frame. Because guess what, I'm not gonna be able to reduce any cycle climb if it takes me three weeks in testing the application. It simply can't be done. So if you're asking me to deliver a change in 10 minutes, it can't be done. It's three weeks plus 10 minutes. If on the other hand I have an automated testing suite, then it can be 10 minutes, or whatever the lag time of deployment all the way into production is. So practices and appropriate tools are absolute. [Agile Coach, Org A]

Lack of appropriate collaboration tools was also seen as a major hindrance to sustained usage – this was discussed to a greater extent in the focus group which was summarised as:

It's funny most of us have the same thing. It's lack of good collaboration tools for dislocated teams. Since agile promotes collaboration and better communication, I think we haven't seen a good tool that we could use, or a technological solution that could solve the problem. [focus group participant]

In summary, based on the evidence from both interview and focus group findings, both *agile practices* and *tool support* were retained in the Technological category.

6.4. Organizational [changed to organizational factors and added organization structure in the refined model]

The two factors in the initial model: (i) *top management support*, and (ii) *methodology champion* found further support in both interview and focus group data.

6.4.1. Top management support

Focus group participants discussed positive aspects of top management, such as effective and lean leadership that can facilitate sustained agile use by establishing appropriate organizational strategies, structures, and processes. Interview data (15 specific citations) shed some valuable insights on the significance of management support, commitment, and participation in implementing change processes related to funding, organizational structure, and culture.

Interview participants identified some key differences between the influence of top management on the sustained usage of prescriptive IS methodologies and on adaptive methodologies such as agile. For example, in an agile context, *top management support* does not merely imply providing training, financial support, and sponsorship, but rather involves active involvement, participation,

and commitment to effectively implement the agile process. This was explained by a developer at Org C:

You've gotta go through it to - participate and understand what's involved. It's just not saying, 'Oh yes agile's really good and it's the right behaviour' - you know it better by actually being involved in agile. Well actually our CIO is the head of IT and other top executives, they're aware, they're certainly aware. And they also talk about our company being an agile organization, but it means something different to them [because they are not directly involved in being agile], for example, responsive to change and that sort of thing.

Adopting a lean leadership mindset, developing a culture of team problem solving, and creating a friendly and non-threatening environment where people feel valued and respected were identified as important characteristics of an agile supportive organization:

A well-run Agile organisation will actually pay attention to make work more enjoyable; make work safer; make work more fun as part of the agile improvement process. It's an approach that essentially says, 'Show me what it is that you're doing. I want to show respect to the things that you're doing'...this notion of showing deep respect for people in order to promote and sustain continuous improvement, that's critical. [Business Analyst, Org B]

I think for agile to survive there has to be a wider acceptance that an agile way of working and thinking is better. For example, the focus on optimizing the outputs of the team rather than the individual, i.e. moving from an individual culture to a team culture has to be recognized by management. [Scrum Master, Org C]

From an organizational perspective it has to be seen that by working more collaboratively and letting go of some of the detailed control is okay, that's okay. [IT Delivery Manager, Org E]

The significance of developing effective strategies aimed at developing a whole system that facilitates agile values and principles via organization-wide changes was deemed critical to lasting transformation. Such strategies include transitioning to a flatter structure with funding and resourcing models that match the agile way of working. The role of *top management support* in establishing various roles such as Scrum Master and Agile Coach was discussed in the focus group:

What really enhanced agile sustainability was management recognizing that in order to get the traction we were looking for in agile and maintain that sustainability was the need to bring in some external resources to help with the organization as a whole and to focus on teams to help teams move forward.

Findings also stressed the significance of establishing a unified focus on anticipating and dealing with any challenges and disruptions in the organization wide-agile implementation:

Some people view agile as a 'fly by the seat of your pants' approach to development when actually it's a very disciplined process. Its capabilities are supported by a number of enablers such as technological, organizational,...; so, a unified approach to sustaining the efforts of such initiatives with a focus on improving the overall system is critical..., it is always more effective when you focus on improving the system as opposed to people working in the system. [Developer, Org D]

6.4.2. Methodology champion

Interview participants provided valuable insights (8 specific citations) on the positive influence a *methodology champion* can have on sustained agile usage. Findings highlighted differences between various agile roles such as methodology champion, agile coach, and

Scrum Master. Whereas a Scrum Master may facilitate the appropriate use of Scrum practices for a team, an agile coach is usually involved in facilitating the effective application of agile practices. The agile coach has a deeper understanding of the social aspects of teams, managing change, and facilitating their self-managing behavior, but the methodology champion role is crucial for the effective agile dissemination throughout the organization. A *methodology champion* operates at a higher level (i.e. organizational) than both the scrum master and agile coach. The champion also acts as an intermediary and liaises between the agile coach and top management. A developer from organization C explained how in spite of having strong support from the product owner and management, there was a lack of understanding at the board level. The methodology champion was recognized as the “key driver whose role was absolutely critical to survive the initiative” (Developer, Org C) who was instrumental in creating a community interest and facilitated a number of learning strategies at the organizational level. A *methodology champion* was perceived as a senior member within the organization whereas an *agile coach* is usually someone external to the organization. In situations where teams with high levels of *agile experience* can become complacent and tend to degrade to previous practices over a period of time, a methodology champion can play a critical role in establishing appropriate coping mechanisms. A Scrum master at Org C explained:

I think this is where the sustainability question is quite important: to maintain clear standards? maintain quality? Because after about a two-and-a-half-year program of work, teams might start degrading. How do you sustain? I think perhaps a championship role could do a whole range of things, establish sustainable training programs, also mentoring programs as well. Someone who has that depth of experience to look at those patterns and dynamics of what happens to teams over time and then what suggestions and solutions might be available to address some of those problems.

Therefore, while *top management support* may support establishing a favorable environment for agile usage, the impetus for its effective and sustained usage comes from individual initiatives of a *methodology champion*, who can play a significant role in encouraging and facilitating the on-going usage of agile practices. Some key responsibilities associated with the role are convincing management, eliminating barriers to successful diffusion and implementation, and mentoring (Conboy and Fitzgerald, 2010), as well as solving problems that hinder agile expansion, implementing change processes that do not fit the agile way of working, and facilitating coordination between various parties that the development depends on (Manen and Vliet, 2014).

6.4.3. Organization structure [added to the refined model]

Another key concept that emerged was the need for an *organization structure* (9 specific interview citations) that supports the agile way of working. Participants explained how their current models, such as funding models, project management structure, and formal reporting systems were against the agile way of working. Interviewees shared examples of organizations using agile methods for more than five years that support a flatter structure and low need for control were still struggling to evolve as successful agile organizations due to strong hierarchical or bureaucratic organizational structures. Interviewees from Org E explained that their funding structure, which allocates funds for achievement of specific final goals, was incompatible with the agile model where the final goal may be uncertain due to its adaptive and flexible processes. The project manager explained how Prince2 would have better supported sustained agile use in their organization:

So the whole concept of Prince2, or event based reporting or event based meetings, would have worked a lot better with an agile set

up than formal standard, monthly steering committee meetings because agile does fit reasonably well with the whole Prince2 thinking. [Project Manager, Org E]

This view was supported by other practitioners who had similar experiences:

Because a lot of organizations are reasonably hierarchical and I think agile tends to support more of a flat structure ...this is what I like about the Prince2: it comes back to a business case and the business value and making sure that the cost of the project and the value that it's bringing to the organization are continually being monitored. And that's what we're continually doing in agile when we're trying to skin stories down so that we deliver the minimum amount of functionality that will give the business value. 'Cause, I mean, the minimum functionality is the minimum cost and so we're continually getting people to think in that space. [Developer, Org C]

In addition, citations that were found more relevant to *organisation structure* (from *compatibility* in the initial Innovation category) were also counted (combined citation count: *compatibility* (4) + *organization structure* (9) = 13). Given this strong evidence from both the focus group and interview data, *organization structure* was added as a factor. Table 5 summarizes the initial and refined factors and includes the citation counts for each factor. All citations come from the analysis of the data used in this study. The citation counts we provide for initial SAUM factors refer to when participants in this data set offered support related to the factors' original meanings included in the initial model. The citation counts we provide for the refined SAUM factors refer to when participants in this data set offered support related to the refined meanings for factors that we revised in the refined SAUM. Where the initial factors were retained in the refined SAUM (e.g. Technological and Organizational factors), we counted all citations under the refined SAUM factors rather than the initial SAUM factors to show that these factors are supported by this data and therefore retained in the refined SAUM.

6.5. Summary of refined sustainability factors

In summary, this data helped us make revisions throughout the four main categories of factors that impact sustained agile usage. The data from both the focus group and interviews indicated that having evidence of appropriate metrics to demonstrate the value of agile methods was more important for sustained use. Therefore, the original factor *relative advantage* was changed to *result demonstrability*. Furthermore, given that agile metrics are evidenced at the team level, *result demonstrability* was moved to Agile Team Factors category (Sociological category in the initial model). The overlap between *compatibility* as an Innovation factor and *Organizational Structure* factor was addressed by removing *compatibility* from the Innovation category and adding a new factor called *organization structure* under the Organizational category. With no factors left, the Innovation category was removed from the model.

The Sociological category was re-specified as Agile Team Factors with four factors: (i) *agile experience* (changed from *experience*), (ii) *agile mindset* (new factor derived from sociological team characteristics), (iii) *agile coach* (added), and (iv) *result demonstrability* (moved from the Innovation category). Based on strong evidence from both interview and focus group findings, both *agile practices* and *tool support* were retained in the Technological category. In the Organizational category, the factors *top management support* and *methodology champion* were retained, and a new factor, *organization structure*, was added. We present the refined model in Fig. 3, and the refined set of factors in each of the four categories are summarized in Table 6.

Table 5

Summary of coding interview data and changes to the categories and factors from the initial to the refined SAUM.

Initial SAUM Factors	Citation Count	Factors Changed to:	Refined SAUM Factors	Citation Count
Innovation			[removed]	
<i>Relative advantage</i>	5		[changed to <i>result demonstrability</i> and moved to <i>Agile Team Factors</i>]	
<i>Compatibility</i> [4 citations overlapped with <i>organisation structure</i>]	7		[removed]	
Sociological			Agile Team Factors	
<i>Experience</i>	4		<i>Agile experience</i> (new) [9 specific citations + 2 citations from <i>technical competence</i>]	11
<i>Attitude</i>	3		<i>Agile mindset</i> (new) [new: 11 specific citations + 3 citations from <i>technical competence</i>]	14
<i>Technical competence</i> [2 overlapped with <i>agile experience</i> and 3 with <i>agile mindset</i>]	5		(removed)	
			<i>Agile coach</i> (new)	12
			<i>Result demonstrability</i> (new)	13
Technological			Technological Factors	
<i>Agile practices</i>			<i>Agile practices</i> (retained)	7
<i>Tool support</i>			<i>Tool support</i> (retained)	13
Organizational			Organizational Factors	
<i>Top management support</i>			<i>Top management support</i> (retained)	15
<i>Methodology champion</i>			<i>Methodology champion</i> (retained)	8
			<i>Organization structure</i> (new: 9 specific citations + 4 from <i>compatibility</i> in the Innovation Category)	13

7. Implications

The aim of this study was to explore critical factors that influence sustained usage of AMs in organizations. To achieve this, we adopted a two-phased research approach consisting of an initial focus group followed by qualitative semi-structured interviews. Building and extending on Senapathi and Srinivasan (2012)'s study, we describe how we refine the initial model in an attempt to derive a more comprehensive model. Below, we discuss the theoretical contributions with implications for both research and practice.

7.1. Implications for research

The main contribution of this study is the development of a more comprehensive, refined SAUM. The combination of a focus group and in-depth semi-structured interviews method has shown to be an effective way of triangulating from multiple sources to allow a comprehensive exploration (Robson, 2011) of the key factors conducive to sustained agile usage. This study offers a

useful complement to the few studies that have examined the long-term acceptance of agile methods, yet their focus has been on only a subset of the factors (e.g. innovation) and their results limited to specific agile methods such as Scrum or Kanban (Ahmad et al., 2016; Overhage and Schlauderer, 2012a, 2012b; Schlauderer et al., 2015). The current study seeks to cover the totality of agile methods rather than on individual agile methods or practices. Our refined model emphasizes the distillation of the important factors collected in our data into three meaningful categories of factors: Agile Team, Technological, and Organizational Factors. It can be used as a reference model to guide future studies to understand sustained usage in different agile domains. For example, empirical studies could be conducted for several specific agile development methods such as Kanban and Lean Development. A suggestion of a future research direction would be for the factors developed in our model to be used in other research methods (e.g. in a survey to achieve statistical generalization).

While this study contributes to a cumulative research tradition by building on the research of (Senapathi and Srinivasan, 2012), it

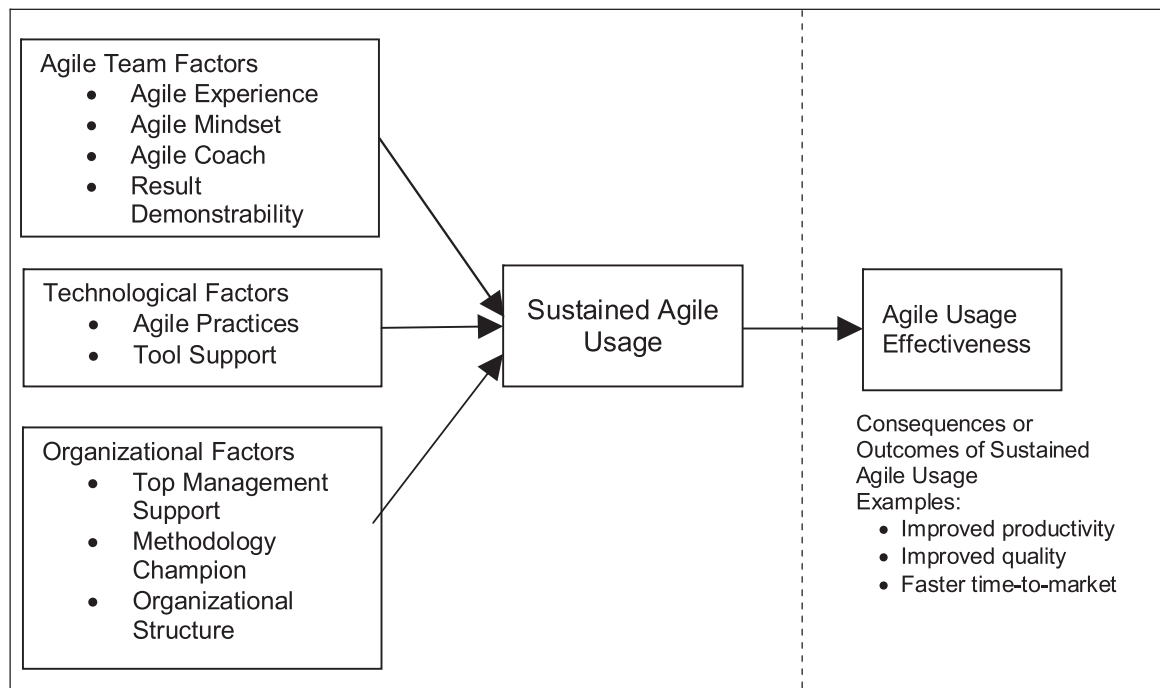


Fig. 3. Refined sustained agile usage model.

also identifies significant ways in which agile differs from other development methodologies with newer meanings and understandings for the factors in the initial model, for example, *relative advantage* and *experience* changed to *result demonstrability* and *agile experience* respectively. We believe this is the first study to examine the role of *coaching* and *agile mindset* in the use of systems development method innovations. Thus, it represents an important step towards elaborating the research agenda to incorporate and further investigate the influence of these novel roles in systems development.

7.2. Implications for practice

Our study, which sought to refine an initial SAUM, is useful in that it identifies a set of the critical factors that influence the effective and sustained use of AMs in organizations. Our findings provide valuable insights that can help agile teams and others (e.g. top management) to better understand and benchmark how agile methods can be effectively sustained in organizations. Here we present some recommendations for practice based on the findings from our study.

Our findings highlight that *result demonstrability*, from a sustained usage perspective, implies evidence of performance improvement (e.g. tracking appropriate metrics) in relation to predecessor methods. Organizations may have a general consensus on the relative benefits of using agile. However, without sound rationale and evidence of improvement metrics, their continued use will be hindered. Therefore, it is strongly recommended that organizations implement specific strategies to track evidence and measures of improvements both from technical (e.g. measuring improvements in technical aspects such as lead time and delivery) and business perspectives (e.g. customer satisfaction, complying market need and expectations). As related research found the importance of project management success factors incorporated into agile iteration objectives (Drury-Grogan, 2014), this could be a plausible way to measure improvements by using agile if organizations use project management success factors.

The absence of an *agile coach* was identified as one of the key factors for agile teams to degrade to their previous traits, leading to the ineffective use of agile practices in many organizations. This implies that organizations must make substantial commitments for the provision of coaching facilities like investing in expert coaches. Also, developing and empowering internal coaching communities is critical to achieving sustainable and lasting change so that organizations can be self-sufficient in sustaining effective agile use.

Another significant finding from this study is the factor *agile mindset* that encapsulates a number of important characteristics. The results stress that it is indispensable that these characteristics exist at higher levels (e.g. at team or organization levels). It is important that organizations identify and implement appropriate strategies to internalize these values at team, department, and organizational levels rather than just implement agile as a process.

The findings also reveal that *top management support*, *organization structure*, and cultural challenges are very real issues facing organizations. This implies that top management does not merely accept the concept of agile but commits to the role they have to play in order to sustain agile in their organizations. In addition to traditional support mechanisms such as financial support and staff training, a key focus should be on developing a whole system that facilitates agile values and principles via organization-wide changes, for example developing effective strategies for establishing dedicated roles (e.g. Agile Coach), creating mentoring programs, transitioning to a flatter structure, and funding and resourcing models that match the agile way of working. Top management should have a unified focus on anticipating and dealing with any challenges and disruptions in the organization-wide implementation.

8. Limitations and conclusions

Finally, this study is not without limitations. Firstly, in contrast to the normal convention of conducting a focus group, this focus group was conducted in a conference setting where a targeted group of experienced practitioners self-selected to participate, which imposed some limitations. For example, it was not

Table 6

Summary of refined sustainability factors.

Innovation Factors (removed from the refined model) - initial model included <i>relative advantage</i> and <i>compatibility</i>	
(1) <i>Relative advantage</i> was changed to <i>result demonstrability</i> and moved to Agile Team Factors.	
(2) <i>Compatibility</i> had overlap between compatibility as an innovation factor and organizational structure. This was addressed by removing compatibility from the Innovation category and adding a new factor called <i>organization structure</i> under the Organizational Factors category.	
Agile Team Factors (Sociological category in the initial model included <i>experience</i> , <i>attitude</i> , and <i>technical competence</i> – the refined model includes <i>agile experience</i> , <i>agile mindset</i> , <i>agile coach</i> , and <i>result demonstrability</i>)	
(1) <i>Agile experience</i> (changed from “ <i>experience</i> ” in the initial model):	Specific experience in using agile methods (i.e. <i>agile experience</i>) rather than general experience is critical in facilitating later stages of agile use as it influences practitioners’ willingness and ability to adapt to changes in values and principles associated with agile.
(2) <i>Agile mindset</i> (changed from “ <i>attitude</i> ” in the initial model):	Agile values and principles held by groups of people to the extent that it becomes a norm or way of life, i.e. internalizing agile mindset values at team, department, and organizational levels rather than just implementing agile as a process, is key to sustainable agile transformation.
(3) <i>Agile coach</i> (added <i>Agile coach</i>):	- Coaching that enables teams to become self-managing and adept in effective application of agile methods facilitates sustained agile usage. - Empowering communities: developing and fostering in-house coaching capabilities and communities sustains agile usage.
(4) <i>Result demonstrability</i> (changed from “ <i>relative advantage</i> ” in the Innovation Factors category from the initial model):	- Evidence of performance improvement (e.g. tracking appropriate metrics) in relation to predecessor methods is more important than perceptions on whether a method is easy to use. - Tracking metrics both from technical (evidence of improvements in lead time, delivery times) and business perspectives (customer satisfaction, complying market need, and expectations) facilitates sustained agile usage.
Technological Factors (<i>agile practices</i> and <i>tool support</i> retained in the refined model)	
(1) Agile practices:	Effective use of specific practices such as automated testing, refactoring, and ease of automation of testing facilitates sustained agile usage.
(2) Tool support:	Lack of appropriate tools and technologies (e.g. tools such as IntelliJ provide support to practices such as refactoring, collaboration tools) hinders sustained agile usage.
Organizational Factors (<i>top management support</i> and <i>methodology champion</i> were retained and <i>organization structure</i> added in the refined model)	
(1) Top management support:	- Lack of a unified approach and overall vision for organizational wide implementation and inability to leverage management support hinders sustained agile usage. - Emphasizing supplementing agile use from a governance perspective, i.e. integrating it throughout the organization, facilitates sustained agile usage.
(2) Methodology champion:	Effective champion at the organizational level eliminates barriers to sustained agile usage.
(3) Organization structure:	- Lack of an organization structure that reflects the values and principles of agile development hinders sustained agile usage. - Incompatibility in resourcing and funding models hinders sustained usage of methods. - Incompatibility with the teams’ preferred way of working, compatibility issues relating to disparity in tools used by different vendor partners, different internal teams, and providing support in divergent platforms hinders sustained usage of methods.

possible to attribute written and verbal data to individual groups members (i.e. effect of personal context on the data could not be considered during the data analysis phase). Likewise, a theoretical framework was provided to the focus group prior to their discussion because we provided participants with the initial SAUM (Senapathi and Srinivasan, 2012), although we deemed this a necessary part of the process so they could respond to the original model developed and provide feedback about what was missing and what could be changed. Therefore, though the findings include this limitation, effort was taken to evaluate the design and conduct of the focus group using prominent frameworks (Merton et al., 1990).

Secondly, a group of 29 software industry agile practitioners is a relatively large focus group compared to a normal group which usually consists of 6 to 12 participants. This was mitigated by dividing the group into smaller groups of 5 to 6 participants and allowed us to have smaller groups focus on each of the factors. We then held a discussion with all 29 participants so that we could incorporate comments from all participants regarding the smaller groups’ discussions.

Thirdly, the scope of the interview phase was also restricted as we interviewed at five New Zealand organizations. Interviews from more organizations and different geographic locations would have strengthened the insights obtained from this phase or could have

provided some new additional insights. Trying to address this limitation by incorporating more organizations for interviews would have added more resource constraints on the one hand, but on the other hand, it was also deemed unnecessary given the role of this phase (i.e. further validation of the research model) in the overall multi-method approach adopted in this study. Because our focus group was conducted in the USA with American employees, we do believe that this mitigated the limitation of New Zealand interviews as our data came from two countries.

In conclusion, we refined the initial SAUM model developed. Our focus group and interview data provided international perspectives and revised the initial factors so that the Innovation category (*relative advantage* and *compatibility*) was removed - *relative advantage* was changed to *result demonstrability* and was moved to Agile Team Factors category. The overlap between *compatibility* as an Innovation factor and Organizational Structure factor was addressed by removing *compatibility* from the Innovation category and adding a new factor called *organization structure* under the Organizational category. The Sociological category was re-specified as Agile Team Factors with four factors: (i) *agile experience*, (ii) *agile mindset*, (iii) *agile coach*, and (iv) *result demonstrability*. The Technological factors *agile practices* and *tool support* were retained in the model. Finally, the Organizational factors of *top management support* and *methodology champion* remained in the model, and or-

ganization structure was added to the model. Future research can now apply these categories and factors in other organizations and countries to assess sustained agile usage. It can also examine the process and outcomes for these factors as we focused primarily on establishing and refining the factors.

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